Lecture 5 - Stakeholder Engagement, Communication and Smart Cities

Lecturer notes, slide by slide

With activities, student questions and model answers

Deck	ASM_Lecture 5_Stakeholder Engagement Communication and Smart Cities_revised.pptx
Slides	29 - numbering matches the revised PowerPoint
Lecturer	Jenan Hussein
Revision	5 October 2026; content and visual approach adapted from approved Lecture 1
Teaching use	Read the summary first, then keep the slide cards beside you. Say gives the explanation; Ask and Tip support delivery.

What changed

Seven focused objectives; clearer links to business decisions; the original six-part sequence, cases and activities; corrected qualifications where benefits or rules were too absolute. Full teaching notes are also embedded in the PowerPoint.

Seminar connection

Seminar 4 (Communicating sustainability to stakeholders and the public). The stakeholder-mapping activity is the direct input for it, and it fills the stakeholder section of the semester paper.

1. The lecture at a glance

The big idea

A company or city project depends on the people it affects. The first half identifies stakeholders and matches the participation promise to real decision-making powers. The second half connects honest claims to evidence and applicable rules, then asks where data can improve GI operations. The recurring city cases add a management question: what decision needs better information, who owns it, and how will the result be evaluated?

Learning outcomes

1. Identify stakeholders affected by a business or GI project.
2. Use power and interest to plan engagement.
3. Choose an appropriate level of participation.
4. Apply stakeholder mapping to a Prague development debate.
5. Write evidence-based claims and recognize greenwashing.
6. Explain how data supports smart GI management.
7. Propose a Prague pilot with an owner, budget and outcome.

Words to explain early

Term	Plain-language meaning
Stakeholder	A person or group that affects, or is affected by, an organisation or project.
Power-interest grid	A planning tool for deciding how to engage stakeholders; positions can change.
IAP2 Spectrum	Inform, consult, involve, collaborate and empower: different levels of public influence and promises.
Greenwashing	A misleading impression about environmental performance, including through vague, selective or unsupported claims.
Recognised excellent environmental performance	A specific legal condition for generic claims under Directive 2024/825; ordinary self-declared evidence is not enough.
Smart GI	Use of data and technology where it improves the management of green infrastructure.
Digital twin	A model used to test scenarios; usefulness depends on data, assumptions and validation.
Data governance	Who owns, maintains, can access and is accountable for data and its use.

Timing plan

These are the original suggested block allocations, retained for planning. Adjust them to your actual teaching slot. Individual slide timings are approximate; do not add them to the block totals.

Block	Slides	Minutes
Welcome, objectives, agenda	1-3	8
Part 1 - Stakeholder theory	4-6	20
Part 2 - Public participation and governance	7-9	20
Part 3 - Prague debates	10-11	10
Activity: map your site's stakeholders	12	25
Break	13	15-20
Part 4 - Communicating sustainability	14-15	15
Part 5 - Greenwashing	16-18	20
Part 6 - Smart cities and smart GI (city tour)	19-25	35
Key messages, reading, questions	26-28	10

Before class

Find one current Prague news story about a park, riverbank or development dispute for slide 11 - there is always one. Bring one advertisement (photo on your phone is enough) with a vague green claim for slide 15. Check that every project group knows its site - slide 12 does not work without it.

Use the revised deck named on this cover. Confirm the timetable, seminar deadline and assessment details with the course organiser. Distinguish classroom assumptions from observed results.

Regulatory references were checked on 5 October 2026. For a real company or advertisement, confirm the applicable national rules and reporting year.

2. Slide-by-slide notes

Slide numbers and titles match the revised deck. These are suggested words and prompts, not a script to read verbatim.

01 Applied Sustainability in Business and Management (1 min)

Say: Welcome. Say: so far we measured things; today we deal with people - the ones who can stop a project or carry it - and with how we talk to them.

02 Learning Objectives (3 min)

Say: Read the seven objectives. Connect each to a decision a manager needs to make. The first half introduces the tools and business problem; the second half applies them to the cases and activities. Students will use the same tools for their Prague semester site.

03 Today's Lecture Agenda (2 min)

Say: Six parts. The activity before the break is long (25 minutes) and important for the project.

04 Part 1 - Stakeholder Theory (1 min)

Say: Announce the part.

05 Who is a Stakeholder? (8 min)

Say: Freeman's definition from 1984: anyone who can affect, or is affected by, what the organisation does. For a green project in a city that means residents and neighbourhood groups, the city and district offices, businesses, developers and investors, NGOs and universities - and future generations, who cannot speak for themselves. Ask students to name stakeholders of a new park in their district; they will forget some, which is the point.

06 The Power-Interest Grid (10 min)

Say: Draw the 2×2 grid on the board. High power and high interest - manage closely: city hall, a major developer. High power, low interest - keep satisfied: a national regulator. Low power, high interest - keep informed and give them a voice: residents, local NGOs. Low power, low interest - just monitor: occasional visitors. The position tells you what to do; that is why the grid is useful. Warn: people move between boxes - residents with low power become high power when they organise.

Ask: Where would you put the local newspaper? (Often high interest, and its power depends on the story.)

07 Part 2 - Public Participation and Governance (1 min)

Say: Announce the part.

08 Levels of Participation (IAP2 Spectrum) (8 min)

Say: These five levels are the IAP2 Spectrum, distinct from Arnstein's eight-rung ladder. Inform, consult, involve, collaborate and empower describe different promises to the public. Match the promise to the legal decision-maker and the project. Do not promise a shared decision if the sponsor will only collect opinions.

Ask: Which level is feasible for this company project, and how would you communicate that?

Source references: see the source list at the end of this guide.

09 Why Participation Improves Projects (8 min)

Say: Participation can reveal local knowledge, expose concerns and support long-term care. It can reduce avoidable delays, but cannot guarantee agreement or prevent legal challenge. Treat it as part of risk management and a fair decision process. Ask what the project sponsor must do with the feedback received.

10 Part 3 - Prague Debates (1 min)

Say: Announce the part.

11 Prague: Where Interests Collide (8 min)

Say: Three recurring Prague debates: building near parks and the river, protecting green wedges from construction, and riverbank redevelopment - recreation versus commercial use. Use the current news story you brought. Ask students to name the stakeholders in it and put two of them on the grid. Observation for them: every public fight is a stakeholder map waiting to be drawn.

12 Activity: Map Your Site's Stakeholders (25 min)

Say: Project groups list every stakeholder of their semester site, place them on the grid, and for the top three write what they want, what they fear and how to engage them. This goes straight into Seminar 4 and the paper. Details in section 3.

13 Break (15-20 min)

Say: After the break: how to talk about sustainability without lying, and what smart cities do with data.

14 Part 4 - Communicating Sustainability (1 min)

Say: Announce the part.

15 Principles of Credible Communication (12 min)

Say: A credible claim states what changed, where, over which period and using what evidence. Include limits and missed targets. Tailor the explanation to the audience without changing the facts. Any numerical wording used in this exercise is fictional until supported by real data.

Ask: Take a claim from an advertisement and identify the missing evidence.

16 Part 5 - Greenwashing (1 min)

Say: Announce the part.

17 The Seven Sins of Greenwashing (8 min)

Say: Use TerraChoice's seven sins as a teaching checklist, not a legal test. Explain hidden trade-off, no proof, vagueness, irrelevance, lesser of two evils, fibbing and false labels with clearly hypothetical examples. The actual legal requirements come from the applicable rules discussed next.

18 Environmental Claims and EU Rules (10 min)

Say: Directive (EU) 2024/825 tightens consumer-facing claims. Generic claims need recognized excellent environmental performance relevant to the claim; an arbitrary company document is not enough. It also prohibits claims that a product has neutral, reduced or positive greenhouse-gas impact when based on offsetting. Sustainability labels must meet the directive's conditions. National measures apply from 27 September 2026. Keep this distinct from CSRD reporting and from the separate Green Claims proposal.

Tip: Check national implementation for a real advertisement or penalty question.

Source references: see the source list at the end of this guide.

19 Part 6 - Smart Cities and Smart GI (1 min)

Say: Announce the part.

20 What Makes a City 'Smart'? (6 min)

Say: A smart city uses data and technology to run its services better. Smart Green Infrastructure means: sensors measure soil moisture, temperature and air quality live; irrigation runs only when needed; digital twins simulate floods, heat and planting before money is spent; open dashboards let citizens see the results and trust them. Connect to Lecture 4: this is where the KPIs get their data.

21 Singapore: Nature and Building Operations (4 min)

Say: Use the documented site functions at Gardens by the Bay. Some Supertrees harvest solar energy; others support conservatory ventilation. This demonstrates integration at a particular site, not an island-wide smart-irrigation system.

Ask: For a Prague hotel or office, would the first priority be water use, cooling or maintenance, and what data would help?

Source references: see the source list at the end of this guide.

22 Copenhagen: Managing Cloudburst Risk (4 min)

Say: Revisit Copenhagen through the management and data perspective. Flood modelling helps decide where water should move or be stored. The harbour-bath photograph illustrates waterfront public space, not a particular cloudburst asset. Avoid promising that surface measures are always cheaper than sewers; compare alternatives for the same service level.

23 Rotterdam and Barcelona: Decisions to Measure (5 min)

Say: Keep both cases, but use this return to ask about decisions and measurement. Rotterdam illustrates shared space and water management; Barcelona illustrates traffic and public-space change. Claims about sales, pollution or walking need a named evaluation and location.

Ask: What baseline would nearby businesses and residents accept?

24 Vienna - and Prague's Opportunity (6 min)

Say: Vienna’s Danube Island shows multiple uses of a green asset. Prague has natural assets and city plans, but do not assume it lacks digital systems. Students should first inspect existing data, responsible teams and operating problems, then propose a pilot only where it fills an evidenced gap.

Ask: Which decision would change because of the proposed sensor or dashboard?

25 The Technology Stack (5 min)

Say: Four layers, from maps to decisions. GIS: where is the green, who can reach it. Sensors: how is it performing right now. Digital twins: what happens if we change it. Dashboards: is the strategy working. Students only need to know the four layers and what question each answers.

26 Key Messages (4 min)

Say: Work through the eight messages. Ask students to connect each to a decision, example or tool from this lecture. Distinguish potential benefits from measured results.

27 Recommended Reading (2 min)

Say: Point to the Seminar 4 preview: communicating sustainability for their site's stakeholders. Next lecture is the last one: ESG, regulation and the future of urban development.

28 Any questions? (5 min)

Say: Take questions.

29 Image credits

Say: No need to show.

3. How to run the class activities

Map Your Site's Stakeholders (slide 12, 25 minutes)

1. Project groups sit together with a large sheet of paper. Draw the power-interest grid.

2. 5 minutes: list every stakeholder of the site - residents, businesses, the district office, city hall, IPR Prague, tourists, NGOs, schools, transport company, developers, and anyone else.

3. 7 minutes: place each one on the grid. Discuss disagreements - they usually reveal something.

4. 8 minutes: for the top three (highest power and interest) write: what do they want, what do they fear, and how will we engage them (meeting, survey, co-design workshop, formal consultation).

5. 5 minutes: two groups present their top three. Ask the room: did they miss a stakeholder?

6. Tell students to photograph the sheet - they need it for Seminar 4.

Watch for: Groups put 'the city' as one stakeholder. Split it: the district office, city hall, IPR Prague and the transport company often want different things.

Business decision prompt

For each proposal, ask students to name the payer, operating owner, affected stakeholders and one measurable outcome. Use assumptions openly when data is unavailable.

4. If a student asks

Isn't participation just slow?

Participation takes time and resources early, but can expose problems and reduce avoidable delay. It does not guarantee agreement or remove formal appeal rights. Choose an approach proportionate to the decision.

Can a company say 'sustainable' at all now?

Use precise claims with evidence. Generic environmental wording may face stricter legal conditions, so replacing it with a number is not automatically enough either. Check scope, relevance, baseline, verification and national rules.

Is the Green Claims Directive law or not?

Keep the separate Green Claims proposal distinct from enacted law. This lecture relies on Directive 2024/825 and its applicable national measures, not on assuming the proposal became law. Check the current legislative record before making a claim about the proposal’s status.

Do smart cities collect data about people?

Smart Green Infrastructure mostly measures the environment - soil, temperature, water - not people. Where cities count visitors or traffic, data protection rules apply. It is a fair question for the stakeholder map: who owns the data?

5. Questions with model answers

Use these for discussion, homework or test preparation. Model answers are short examples; other well-supported answers can also be valid.

1. Give Freeman's definition of a stakeholder and list six stakeholders of a riverbank park in Prague.

Recall

Model answer: Any group or individual who can affect, or is affected by, the achievement of the organisation's objectives. Examples: nearby residents, the district office, city hall / IPR Prague, cafés and boat operators, tourists, environmental NGOs, the river authority, future generations.

2. Draw the power-interest grid, name the four strategies, and give one example stakeholder for each box.

Understand

Model answer: High power / high interest: manage closely (city hall, major developer). High power / low interest: keep satisfied (national regulator). Low power / high interest: keep informed and give a voice (residents, local NGOs). Low power / low interest: monitor (occasional visitors).

3. List the five levels of public participation from weakest to strongest and explain the difference between consultation and collaboration.

Recall

Model answer: The IAP2 Spectrum comprises inform, consult, involve, collaborate and empower. Consultation gathers input while the responsible body retains the decision. Collaboration involves partnering in developing alternatives and the preferred solution. State the actual decision rights; this is distinct from Arnstein’s eight-rung ladder.

4. Give three reasons why involving stakeholders early makes a project better, and one cost of skipping it.

Understand

Model answer: Participation can reveal local knowledge, expose concerns early and support long-term care. It provides a transparent basis for decisions, but does not guarantee agreement or prevent legal challenge. Ignoring stakeholders can produce delay and conflict.

5. What are the four principles of credible sustainability communication? Rewrite the claim 'our hotel is eco-friendly' so that it follows them.

Apply

Model answer: Use specific scope and metrics, attached evidence, a complete account including limitations, and language suitable for the audience. Fictional exercise example: "Our hotel reduced energy per guest night by 30% against 2021 under the same boundary; water use remains above target." This sentence is usable only if data and methodology support it; do not invent verification.

6. Name the seven sins of greenwashing and give a real or invented example for three of them.

Recall

Model answer: Hidden trade-off, no proof, vagueness, irrelevance, lesser of two evils, fibbing, false labels. Examples: 'recycled paper' from a factory with high emissions (hidden trade-off); 'natural' shampoo (vagueness); 'CFC-free' spray when CFCs are banned anyway (irrelevance).

7. What does EU Directive 2024/825 change for companies from September 2026?

Understand

Model answer: National measures implementing Directive 2024/825 apply from 27 September 2026. Generic environmental claims require recognised excellent environmental performance relevant to the claim. Product greenhouse-gas claims based on offsetting are prohibited. Sustainability labels must meet the directive’s conditions. Check the precise claim and national rules.

8. What makes a city 'smart'? Describe four things smart Green Infrastructure does.

Recall

Model answer: A city that uses data and technology to improve services, sustainability and quality of life. Smart GI: sensors monitor soil moisture, temperature and air quality; irrigation runs only when needed; digital twins simulate floods and heat before investing; open dashboards show results to citizens.

9. Compare Copenhagen's cloudburst plan and Barcelona's superblocks: what problem does each solve, and what do they have in common?

Understand

Model answer: Copenhagen addresses extreme rainfall through routes, storage and drainage. Barcelona’s superblocks address traffic and public-space use in selected areas. Both invite evaluation of local outcomes and trade-offs. Specific cost, pollution or business effects need a named study, location and period.

10. Propose three first steps for a smart GI programme in Prague after checking existing systems. Name the responsible stakeholder for each.

Apply

Model answer: First inspect existing systems and the operating problem. A proposed pilot could add soil-moisture monitoring with a parks manager, map heat and access with the relevant city data team, or report agreed KPIs with a named owner. Identify budget, data quality, maintenance and the decision each pilot supports.

Handout version: questions only

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2. Draw the power-interest grid, name the four strategies, and give one example stakeholder for each box.

3. List the five levels of public participation from weakest to strongest and explain the difference between consultation and collaboration.

4. Give three reasons why involving stakeholders early makes a project better, and one cost of skipping it.

5. What are the four principles of credible sustainability communication? Rewrite the claim 'our hotel is eco-friendly' so that it follows them.

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Source list for revised passages

Checked 5 October 2026. These sources support the revised passages; the original recommended reading and image credits remain in the deck. Company material is identified as company-reported evidence.

1. www.iap2.org - slides 8

https://www.iap2.org/page/pillars

2. eur-lex.europa.eu - slides 18

https://eur-lex.europa.eu/eli/dir/2024/825/oj/eng

3. www.gardensbythebay.com.sg - slides 21

https://www.gardensbythebay.com.sg/en/the-gardens/sustainability-efforts.html